Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)			**Any 3 × (1) from:** Particles vibrate (1) More **or** faster (1) collide with neighbours **or** pass on energy to neighbours (1) Free electrons (1) [free] electrons increase in kinetic energy **or** move faster (1) Electrons pass on the energy as they move through the lattice **or** electrons collide with the lattice owtte (1)	3			3		
	(b)			**Any 2 × (1) from:** Convection [currents] (1) Heated liquid rises [from solar panel] (1) Hot liquids are less dense (1) Accept cold liquids fall (1) and cold liquids are more dense (1)		2		2		
	(c)			$960 \times 2 = 1\,920$ (1) $1\,920$ **(ecf)** $\times \frac{1}{3} = 640$ [J] (1) **Alternative:** $\frac{960}{3} = 320$ (1) 320 **ecf** $\times 2 = 640$ [J] (1) Award 1 mark for answer of 576 [J]		2		2	2	
				Question 4 total	**3**	**4**	**0**	**7**	**2**	**0**